
Derivation of spinDMFT and extension to finite temperatures

Timo Gräßer
Condensed Matter Physics
Uhrig Group

Recap of spinDMFT¹

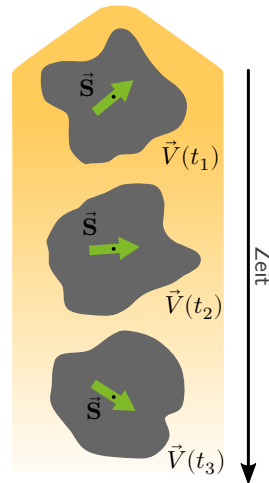
- approximation for dense spin systems at infinite temperature
- local spin environments are replaced by dynamic Gaussian mean-fields
- single-site problem

$$\mathbf{H}^{\text{mf}}(t) = \vec{V}(t) \cdot \vec{\mathbf{S}}$$

- self-consistency condition (Heisenberg model)

$$\overline{V^\alpha(t)V^\alpha(0)}^{\text{mf}} = J_Q^2 \langle \mathbf{S}^\alpha(t) \mathbf{S}^\alpha(0) \rangle = J_Q^2 \int D\mathcal{V} p(\mathcal{V}) \frac{\text{Tr}\{\mathbf{S}^\alpha(t) \mathbf{S}^\alpha(0)\}}{2S+1}$$

- small numerical effort, possibility for extensions



¹T. Gräber et al., Phys. Rev. Res. **3**, 043168 (2021).

Argumentative derivation

- coordination number z
- Frobenius norm of mean-field commutators are suppressed $\rightarrow$ they behave classically

$$\frac{\|[\mathbf{V}^\alpha, \mathbf{V}^\beta]\|_{\text{F}}}{\|\mathbf{V}^\alpha \mathbf{V}^\beta\|_{\text{F}}} \propto \frac{1}{z}$$

- spins are only weakly correlated

$$\langle \mathbf{S}_i^\alpha(t) \mathbf{S}_j^\beta(0) \rangle \propto \frac{1}{z^{\lambda(i,j)}}, \quad \lambda(i,j) : \text{taxicab distance}$$

- local spin environments: large number of independent random variables $\rightarrow$ central limit theorem $\rightarrow$ dynamics Gaussian mean-fields

The setup

- Heisenberg model, $J_{ij} = J_{ji}$ and $J_{ii} = 0$

$$\mathbf{H} = \frac{1}{2} \sum_{i,j} J_{ij} \vec{\mathbf{S}}_i \cdot \vec{\mathbf{S}}_j$$

- separation of the Hamiltonian with respect to a single spin $\vec{\mathbf{S}}_0$

$$\mathbf{H} = \underbrace{\vec{\mathbf{S}}_0 \cdot \vec{\mathbf{V}}_0}_{\mathbf{H}_0} + \frac{1}{2} \underbrace{\sum_{i,j \neq 0} J_{ij} \vec{\mathbf{S}}_i \cdot \vec{\mathbf{S}}_j}_{\mathbf{W}}$$

- local-environment field

$$\vec{\mathbf{V}}_0 = \sum_j J_{0j} \vec{\mathbf{S}}_j$$

- aim: derivation of the mean-field approximation of the spin autocorrelation function

$$g^{\alpha\beta}(t, 0) := \langle \mathbf{S}_0^\alpha(t) \mathbf{S}_0^\beta(0) \rangle = \frac{1}{d} \text{Tr} \{ \mathbf{U}^\dagger \mathbf{S}_0^\alpha \mathbf{U} \mathbf{S}_0^\beta \}$$

Trotterization (1)

- Trotter's formula

$$e^{\mathbf{A}+\mathbf{B}} = \lim_{n \rightarrow \infty} \left(e^{\frac{\mathbf{A}}{n}} e^{\frac{\mathbf{B}}{n}} \right)^n$$

- applied to the time propagator

$$\mathbf{U} = e^{-i\mathbf{H}t} = \lim_{n \rightarrow \infty} \left(e^{-i\mathbf{H}_0 \frac{t}{n}} e^{-i\mathbf{W} \frac{t}{n}} \right)^n$$

- insertion of identities

$$\mathbf{1} = e^{-i\mathbf{W}t \frac{k}{n}} e^{i\mathbf{W}t \frac{k}{n}}$$

leads to

$$\begin{aligned} \mathbf{U} &= \lim_{n \rightarrow \infty} e^{-i\mathbf{W}t} \prod_{k=1}^n e^{i\mathbf{W}t \frac{k}{n}} e^{-i\mathbf{H}_0 \frac{t}{n}} e^{-i\mathbf{W}t \frac{k}{n}} \\ &= e^{-i\mathbf{W}t} \lim_{n \rightarrow \infty} \prod_{k=1}^n e^{-i\vec{\mathbf{S}}_0 \cdot \vec{\mathbf{V}}_0 \left(\frac{kt}{n} \right) \frac{t}{n}} \end{aligned}$$

Trotterization (2)

- time-evolved environments

$$\tilde{\mathbf{V}}_0(t) := e^{i\mathbf{W}t}\mathbf{V}_0e^{-i\mathbf{W}t}$$

- propagator expressed via the time-ordering operator

$$\mathbf{U} = e^{-i\mathbf{W}t}\mathcal{T}e^{-i\int_0^t \tilde{\mathbf{S}}_0 \cdot \tilde{\mathbf{V}}_0(t')dt'}$$

- remaining expression (still exact!)

$$\begin{aligned} g^{\alpha\beta}(t, 0) &= \frac{1}{d} \text{Tr} \left\{ \overline{\mathcal{T}} e^{i\int_0^t \tilde{\mathbf{S}}_0 \cdot \tilde{\mathbf{V}}_0(t')dt'} e^{i\mathbf{W}t} \mathbf{S}_0^\alpha e^{-i\mathbf{W}t} \mathcal{T} e^{-i\int_0^t \tilde{\mathbf{S}}_0 \cdot \tilde{\mathbf{V}}_0(t')dt'} \mathbf{S}_0^\beta \right\} \\ &= \frac{1}{d} \text{Tr} \left\{ \overline{\mathcal{T}} e^{i\int_0^t \tilde{\mathbf{S}}_0 \cdot \tilde{\mathbf{V}}_0(t')dt'} \mathbf{S}_0^\alpha \mathcal{T} e^{-i\int_0^t \tilde{\mathbf{S}}_0 \cdot \tilde{\mathbf{V}}_0(t')dt'} \mathbf{S}_0^\beta \right\} \end{aligned}$$

Approximating the trace (1)

- introduce basis of states, $|m_0\rangle$ refers to the local site 0 and $|W\rangle$ refers to the rest of the system

$$|\psi\rangle := |m_0\rangle |W\rangle,$$

- expressing the trace

$$\begin{aligned} g^{\alpha\beta}(t, 0) &= \frac{1}{d} \sum_{m_0} \sum_W \langle W | \langle m_0 | \overline{\mathcal{T}} e^{i \int_0^t \vec{S}_0 \cdot \vec{V}_0(t) dt'} \mathbf{S}_0^\alpha \mathcal{T} e^{-i \int_0^t \vec{S}_0 \cdot \vec{V}_0(t) dt'} \mathbf{S}_0^\beta | m_0 \rangle | W \rangle \\ &= \sum_{m,n} \frac{i^{m-n}}{m!n!} \int_0^t dt_1 \dots dt_m dt'_1 \dots dt'_n \sum_{m_0} \sum_W \frac{1}{d} \langle W | \langle m_0 | \\ &\quad \overline{\mathcal{T}} \left[\vec{S}_0 \cdot \vec{V}_0(t_1) \dots \vec{S}_0 \cdot \vec{V}_0(t_m) \right] \mathbf{S}_0^\alpha \mathcal{T} \left[\vec{S}_0 \cdot \vec{V}_0(t'_1) \dots \vec{S}_0 \cdot \vec{V}_0(t'_n) \right] \mathbf{S}_0^\beta \\ &\quad | m_0 \rangle | W \rangle \end{aligned}$$

Approximating the trace (2)

- separate the local from the environmental part

$$\begin{aligned}
 f^{mn}(t_1, \dots, t_m, t'_1, \dots, t'_n) &= \sum_{\alpha_1 \dots \alpha_m} \sum_{\alpha'_1 \dots \alpha'_n} \\
 &\sum_{m_0} \frac{1}{d_0} \langle m_0 | \overline{\mathcal{P}}_{t_1 \dots t_m}^{\alpha_1 \dots \alpha_m} \left[\prod_{p=1}^m \mathbf{s}_0^{\alpha_p} \right] \mathbf{s}_0^\alpha \mathcal{P}_{t'_1 \dots t'_n}^{\alpha'_1 \dots \alpha'_n} \left[\prod_{q=1}^n \mathbf{s}_0^{\alpha'_q} \right] \mathbf{s}_0^\beta | m_0 \rangle \\
 &\sum_W \frac{d_0}{d} \langle W | \overline{\mathcal{T}} \left[\prod_{p=1}^m \tilde{\mathbf{v}}_0^{\alpha_p}(t_p) \right] \mathcal{T} \left[\prod_{q=1}^n \tilde{\mathbf{v}}_0^{\alpha_q}(t_q) \right] | W \rangle
 \end{aligned}$$

- environment trace

$$w^{mn}(t_1, \dots, t_m, t'_1, \dots, t'_n) = \frac{d_0}{d} \text{Tr}_{/\{0\}} \left\{ \mathcal{T} \left[\prod_{p=1}^m \tilde{\mathbf{v}}_0^{\alpha_p}(t_p) \right] \mathcal{T} \left[\prod_{q=1}^n \tilde{\mathbf{v}}_0^{\alpha_q}(t_q) \right] \right\}$$

- cavity trick¹: back actions of a single spin are negligible, replace w^{mn} by a trace over the complete system

¹A. Georges et al., Rev. Mod. Phys. **68**, 13 (1996).

Approximating the trace (3)

- central assumption: the local environment operator $\vec{V}_0$ approximately fulfills Wick's theorem if z is large¹

$$\langle \mathbf{V}_0^\alpha \mathbf{V}_0^\beta \mathbf{V}_0^\gamma \mathbf{V}_0^\rho \rangle = \langle \mathbf{V}_0^\alpha \mathbf{V}_0^\beta \rangle \langle \mathbf{V}_0^\gamma \mathbf{V}_0^\rho \rangle + \langle \mathbf{V}_0^\alpha \mathbf{V}_0^\gamma \rangle \langle \mathbf{V}_0^\beta \mathbf{V}_0^\rho \rangle + \langle \mathbf{V}_0^\alpha \mathbf{V}_0^\rho \rangle \langle \mathbf{V}_0^\beta \mathbf{V}_0^\gamma \rangle + \mathcal{O}\left(\frac{1}{z}\right)$$

- can be shown $\forall t_i = 0$, but a proof for $t_i \neq 0$ is difficult
- strong arguments from the standard spinDMFT derivation
 - components of the environment operator approximately commute with one another
 - environment consists of a large number of essentially independent spins (central limit theorem)
- Wick reduction leads to

$$w^{mn}(t_1, \dots, t_m, t'_1, \dots, t'_n) \approx \sum_C \prod_{P \in C} \langle \mathbf{V}_0^{\alpha_{P(1)}}(t_{P(1)}) \mathbf{V}_0^{\alpha_{P(2)}}(t_{P(2)}) \rangle$$

¹V. E. Zobov, Theo. & Math. Phys. **77**, 1573 (1988).

Auxiliary model (1)

- auxiliary model: single spin coupling to a classical Gaussian dynamic field

$$\mathbf{H}^{\text{mf}}(t) = \vec{\mathbf{S}}_0 \cdot \vec{V}_0(t)$$

- infinite-temperature two-point correlation function

$$g^{\alpha\beta}(t, 0) = \int \mathcal{D}\mathcal{V}_0 p(\mathcal{V}_0) \frac{1}{d_0} \text{Tr} \left\{ \overline{\mathcal{T}} e^{i \int_0^t \vec{\mathbf{S}}_0 \cdot \vec{V}_0(t') dt'} \mathbf{S}_0^\alpha \mathcal{T} e^{-i \int_0^t \vec{\mathbf{S}}_0 \cdot \vec{V}_0(t') dt'} \mathbf{S}_0^\beta \right\}$$

- exactly the same as the previous expression except for

$$\tilde{w}^{mn}(t_1, \dots, t_m, t'_1, \dots, t'_n) = \sum_C \prod_{P \in C} \int \mathcal{D}\mathcal{V}_0 p(\mathcal{V}_0) V_0^{\alpha_{P(1)}}(t_{P(1)}) V_0^{\alpha_{P(2)}}(t_{P(2)})$$

Auxiliary model (2)

- the auxiliary model approximately captures the correct spin dynamics if

$$\int D\mathcal{V}_0 p(\mathcal{V}_0) V_0^\alpha(t_1) V_0^\beta(t_2) = \langle \mathbf{V}_0^\alpha(t_1) \mathbf{V}_0^\beta(t_2) \rangle$$

- self-consistency condition (pair correlations neglected)

$$\int D\mathcal{V}_0 p(\mathcal{V}_0) V_0^\alpha(t_1) V_0^\beta(t_2) = J_Q^2 \langle \mathbf{S}_0^\alpha(t_1) \mathbf{S}_0^\beta(t_2) \rangle$$

Thermodynamic autocorrelation

- the auxiliary model approximately captures the correct spin dynamics if

$$g^{\alpha\beta}(t, 0) := \langle \mathbf{S}_0^\alpha(t) \mathbf{S}_0^\beta(0) \rangle = \frac{1}{Z} \text{Tr} \{ \rho \mathbf{U}^\dagger \mathbf{S}_0^\alpha \mathbf{U} \mathbf{S}_0^\beta \}$$

- density operator

$$\rho = e^{-\beta \mathbf{H}}, \quad Z = \text{Tr} \{ \rho \}$$

- thermodynamic autocorrelation (imaginary time $\tau \in \mathbb{R}$)

$$g_{\text{thd}}^{\alpha\beta}(\tau, 0) := \langle \mathbf{S}_0^\alpha(\tau) \mathbf{S}_0^\beta(0) \rangle = \frac{1}{Z} \text{Tr} \{ e^{-\beta \mathbf{H}} e^{\tau \mathbf{H}} \mathbf{S}_0^\alpha e^{-\tau \mathbf{H}} \mathbf{S}_0^\beta \} = \frac{1}{Z} \text{Tr} \{ e^{(\tau - \beta) \mathbf{H}} \mathbf{S}_0^\alpha e^{-\tau \mathbf{H}} \mathbf{S}_0^\beta \}$$

- relation to real-time autocorrelation through analytic continuation

Trotterization, cavity method and Wick's theorem

- trotterization works the same as for infinite temperature

$$g_{\text{thd}}^{\alpha\beta}(\tau, 0) = \frac{1}{Z} \text{Tr} \left\{ e^{-\beta \mathbf{W}} \mathcal{T} e^{-\int_{\tau}^{\beta} \vec{\mathbf{S}}_0 \cdot \vec{\mathbf{V}}_0(\tau') d\tau'} \mathbf{S}_0^{\alpha} \mathcal{T} e^{-\int_0^{\tau} \vec{\mathbf{S}}_0 \cdot \vec{\mathbf{V}}_0(\tau') d\tau'} \mathbf{S}_0^{\beta} \right\}$$

- replace cavity system by full system

$$w^{mn}(\tau_1, \dots, \tau_m, \tau'_1, \dots, \tau'_n) \approx \frac{1}{Z} \text{Tr} \left\{ e^{-\beta \mathbf{H}} \mathcal{T} \left[\prod_{p=1}^m \mathbf{V}_0^{\alpha_p}(\tau_p) \prod_{q=1}^n \mathbf{V}_0^{\alpha_q}(\tau_q) \right] \right\}$$

- apply Wick's theorem

$$w^{mn}(\tau_1, \dots, \tau_m, \tau'_1, \dots, \tau'_n) = \sum_C \prod_{P \in C} \langle \mathcal{T} \mathbf{V}_0^{\alpha_{P(1)}}(\tau_{P(1)}) \mathbf{V}_0^{\alpha_{P(2)}}(\tau_{P(2)}) \rangle$$

Auxiliary model

- single-site model with dynamic Gaussian mean-field

$$\mathbf{H}^{\text{mf}}(\tau) = \vec{\mathbf{S}}_0 \cdot \vec{V}_0(\tau)$$

- thermodynamic autocorrelation

$$g^{\alpha\beta}(\tau, 0) = \int \mathcal{D}\mathcal{V}_0 p(\mathcal{V}_0) \frac{1}{Z_0} \text{Tr} \left\{ \mathcal{T} e^{-\int_{\tau}^{\beta} \vec{\mathbf{S}}_0 \cdot \vec{V}_0(\tau') d\tau'} \mathbf{S}_0^{\alpha} e^{-\int_0^{\tau} \vec{\mathbf{S}}_0 \cdot \vec{V}_0(\tau') d\tau'} \mathbf{S}_0^{\beta} \right\}$$

- self-consistency condition

$$\int \mathcal{D}\mathcal{V}_0 p(\mathcal{V}_0) V_0^{\alpha}(\tau_1) V_0^{\beta}(\tau_2) = \langle \mathcal{T} \mathbf{V}_0^{\alpha}(\tau_1) \mathbf{V}_0^{\beta}(\tau_2) \rangle \approx J_Q^2 \langle \mathcal{T} \mathbf{S}_0^{\alpha}(\tau_1) \mathbf{S}_0^{\beta}(\tau_2) \rangle$$

- local partition function

$$Z_0 = \int \mathcal{D}\mathcal{V}_0 p(\mathcal{V}_0) \text{Tr} \left\{ \mathcal{T} e^{-\int_0^{\beta} \vec{\mathbf{S}}_0 \cdot \vec{V}_0(\tau') d\tau'} \right\}$$

Outlook

- numerical implementation, comparison to other methods
- proof of positivity of the covariance matrix for finite temperature
- rigorous proof of Wick's theorem for $z \rightarrow \infty$
- consideration of a strongly polarized spin system

$$\rho \propto e^{-h \sum_i \mathbf{s}_i^z}$$

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Thanks for listening!